

Exploring Testing from First Principles

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The Tester (Evil)



We over-complicate testing

- Have you seen some of the definitions?
- Have you tried to write your own definition?

Otherwise

- it might appear too simple
- it might appear easy

And if its too easy and simple

- it might be undervalued
- only junior staff get to do it
- we might get paid less

Solution

Cram in Everything

Software Testing is a process whereby we use a software system deployed in multiple configurations, via its GUI, API - and all exposed interfaces - to gather information about its behaviour, flaws and impact on the user and stakeholder needs and requirements; for the purpose of evaluating the quality, and fitness for purpose, in specific operational environments and conditions.

A possible first draft definition

But we wouldn't stop there

- refine our definition
- simplify, chunk up
- delete sections
- change it over time
- communicate it differently for different people

But that's a communication. It isn't what testing 'is'.

Danger - over-complicate

- How do other fields define themselves?
e.g. complicated fields like Physics?

**I should consult a
dictionary or
something.**

What is Physics?

physics

"the branch of science concerned with the nature and properties of matter and energy."

physics

/ˈfɪzɪks/

noun

noun: **physics**

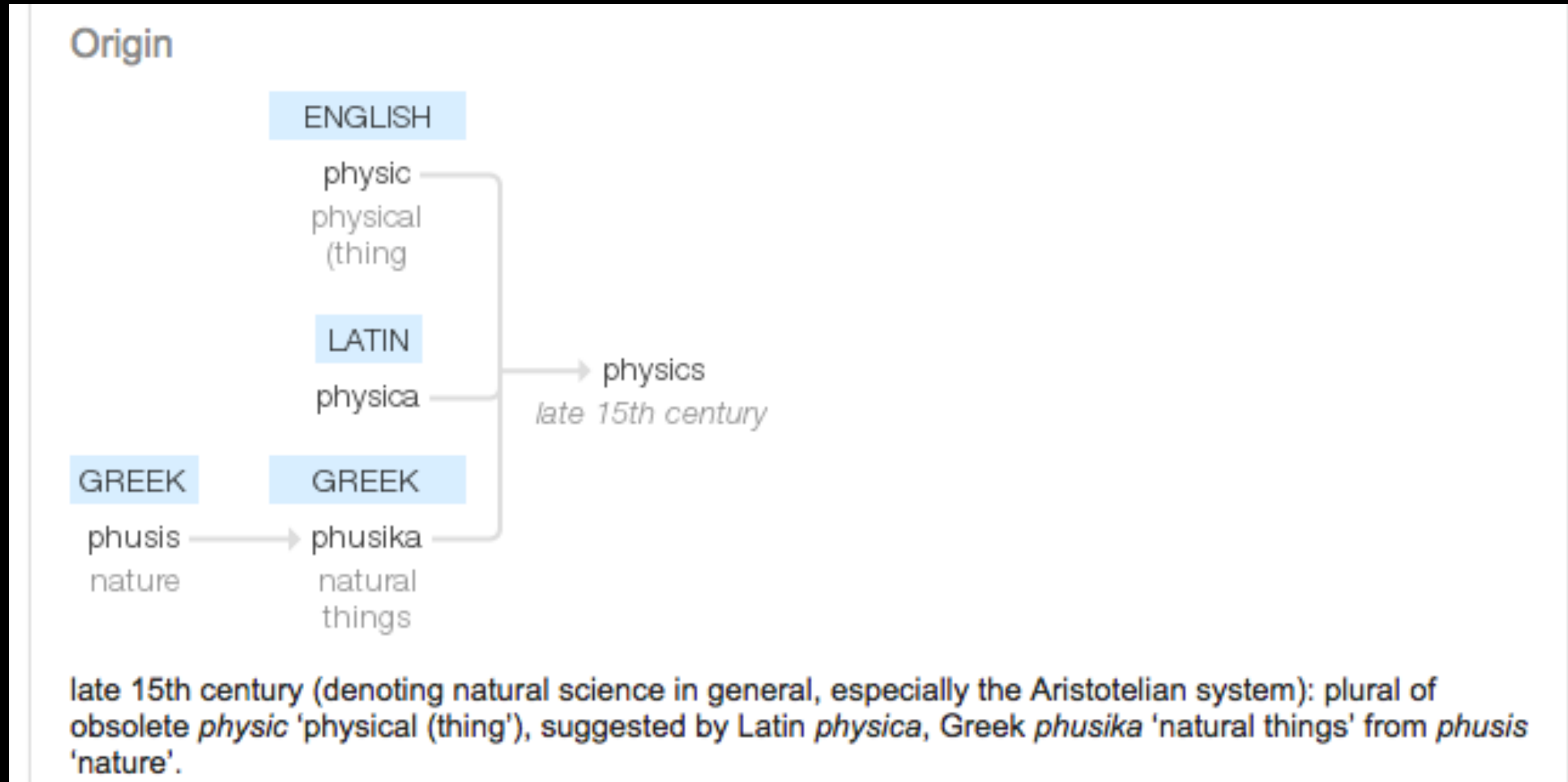
the branch of science concerned with the nature and properties of matter and energy. The subject matter of physics includes mechanics, heat, light and other radiation, sound, electricity, magnetism, and the structure of atoms.

- the physical properties and phenomena of something.

plural noun: **physics**

"the physics of plasmas"

What is Physics?



What about Rocket Science?

What is Rocket Science

Dictionary

🔍

rocket science

noun humorous
noun: **rocket science**

something very difficult to understand.
"we want you to get out and vote—it's not exactly rocket science"

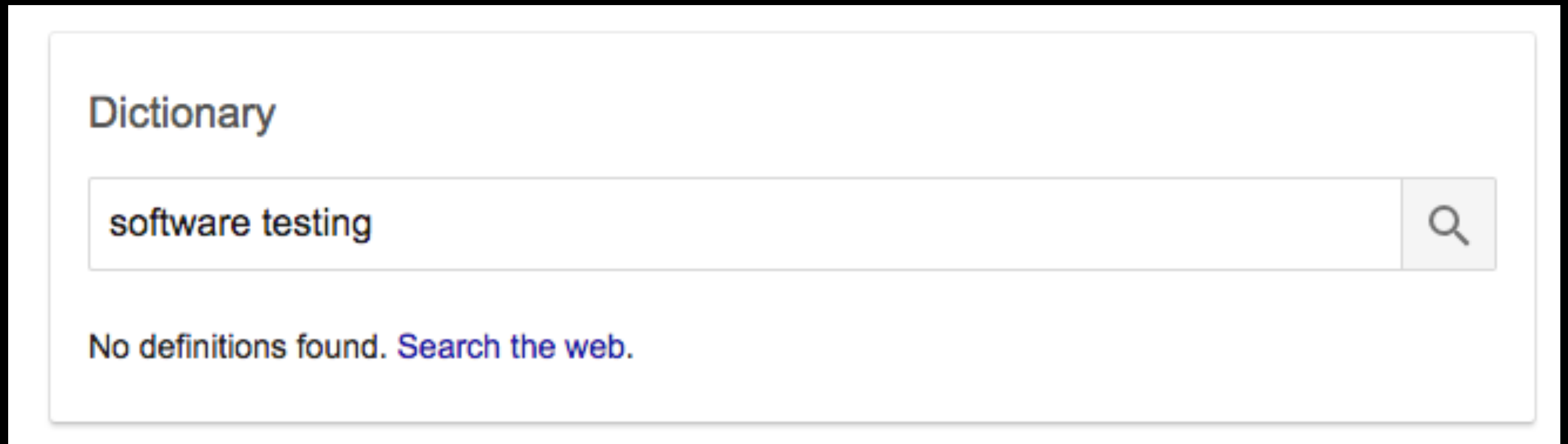
Translate rocket science to

⬆️⬆️

⏴ Show less

Google Dictionary: What is Software Testing?

"No definitions found. Search the web."



We have no choice

We have to make this up for ourselves.

**What is more important?
Imagineedfusticating?**

or

**Working with a good
Imagineedfusticator?**

Imagineedfusticating

"The art of getting exactly what you need when you need it"

*Definition from "The Associated Society of
Imagineedfusticators for Advanced Imagineedfustication"*

An Example

Assume we know nothing about software testing.

"No definitions found. Search the web."

How would we work from first principles?

**Let's use a worked
example**

Pulp App Main Menu

Books

- [Table](#)
- [List](#)
- [FAQs](#)

Authors

- [Authors List](#)
- [Authors FAQ List](#)

Menus

- [Books](#)
- [Authors](#)
- [Publishers](#)
- [Series](#)
- [Years](#)

Search

- [Search Book Titles](#)

Create

- [Create Author](#)
- [Create Series](#)
- [Create Publisher](#)
- [Create Book](#)

**I've been
tasked with
testing this**

I don't want anyone to know that I don't know how to test it, so I nod and then try to figure out what does software testing mean.

So I look for definitions but we've seen how that turned out.

So I go ask google

"What is Software Testing"

What is Software Testing?

Software testing is the process of executing a program or application with the intent of finding bugs in the software. It can also be described as the validation and verification of a software program or application or product or system.

not text from a website that has high rankings for this web search

Validation & Verification

- verification - noun "The process of establishing the truth, accuracy, or validity of something."
- validation - noun - "The action of checking or proving the validity or accuracy of something."

OK, so both are about validity, accuracy, proving and truth.

Those words seem pretty similar, why would I do both?

OK, I'll forget that.

What have I learned?

Software Testing is doing stuff to something to find out stuff like:

- find bugs,
- find things that are wrong,
- find stuff that is right

Warning: Be careful of certitude and absolutes.

Too much theory

I have a job to do

What did they mean when they tasked me with "testing" this.

- find bugs?
- how will I know its a bug?
- anything specific you want me to test it 'for'?
- what do you want to 'do' with what I tell you?
- is testing the input to something else?

"What do you mean?"

OK...

focus on the create author function and make sure it handles a range of input values

"What values?"

You know. Good name. No name. Long names. Errors for bad names. That stuff.

"OK"

"If it creates the user OK then we'll be fine."

We just learned:

- stakeholder management
- needs and requirement analysis
- acceptance condition creation
- risk based testing
- test scope restriction

**Why didn't I ask
"Why do you want me
to test this?"**

**Why? is a belief chain
question**

Why do you want me to test this?

- because its your job
- because I don' t want to test it
- because testing sucks
- because I want proof that it works
- because I don't want the user finding any problems

That might be useful

- we find out the organisation dynamics
- we can 'educate' that testing does not 'prove that it works'
- we could test from a user perspective

The temptation with "Why?" is to chain "Why?"

- why is it my job?
- why don't you want to test it?
- why don't you want the user to find problems?

Unproductive path / conflict.

But I care less about what they believe

I care more about meeting their needs

- What?
- How?
- When?
- Where?

What do I do next?

[Home](#) [Books](#) [Authors](#) [Publishers](#) [Series](#) [Years](#) [Search](#) [Create](#)

Create Author

Enter name of author to create

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Let me plan this out a bit

- No Name - ""
- Good Name - "Bob The Author"
- Bad Name ? "Mr 123"

I can see there some validation when I just hit save

What are the validation rules? I should check those.

Enter name of author to create

Create



Please fill in this field.

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Create Author

Enter name of author to create

Create

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Create Author

Enter name of author to create

Create

Added Author bob the author

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Is that what it is supposed to do?

- How do I know what it is supposed to do?
- How do I know that is the correct message?
- Do I just decide if this works, or are there designs?
- Should it be consistent?
 - "Create" -> "Added"

We just discovered

- A need for a reference standard (oracle)
- conformance to standard
 - requirements testing
 - acceptance condition validation
- The need for exploration: planning, learning, revisiting plan
- Consistency Heuristic

How do I tell people what I have done?

- Better keep some notes to tell people what I've done.
- I need to keep track of new things I want to do e.g. validation rules
- I better keep track of the unanswered questions and concerns I have
- How do people do that?

We just discovered

- Exploratory Testing
- Test Planning
- Test Tools
- Test Reporting

It said it added it, but did it? How can I see it was added?



Home Books Authors Publishers Series

List of Authors

- [Alan Hathway](#) [amend] [x]
- [bob the author](#) [amend] [x]
- [Emile C. Tepperman](#) [amend] [x]
- [Evelyn Coulson](#) [amend] [x]

Can I trust that?

- How is the data stored?
- What data is stored?
- When is the data stored?
- Where is the data stored?
- How can I see the stored data?

We just discovered

- A Testing Attitude Characteristic
 - Doubt
- A Fundamental Testing Technique
 - Questioning
- Create/Read flow

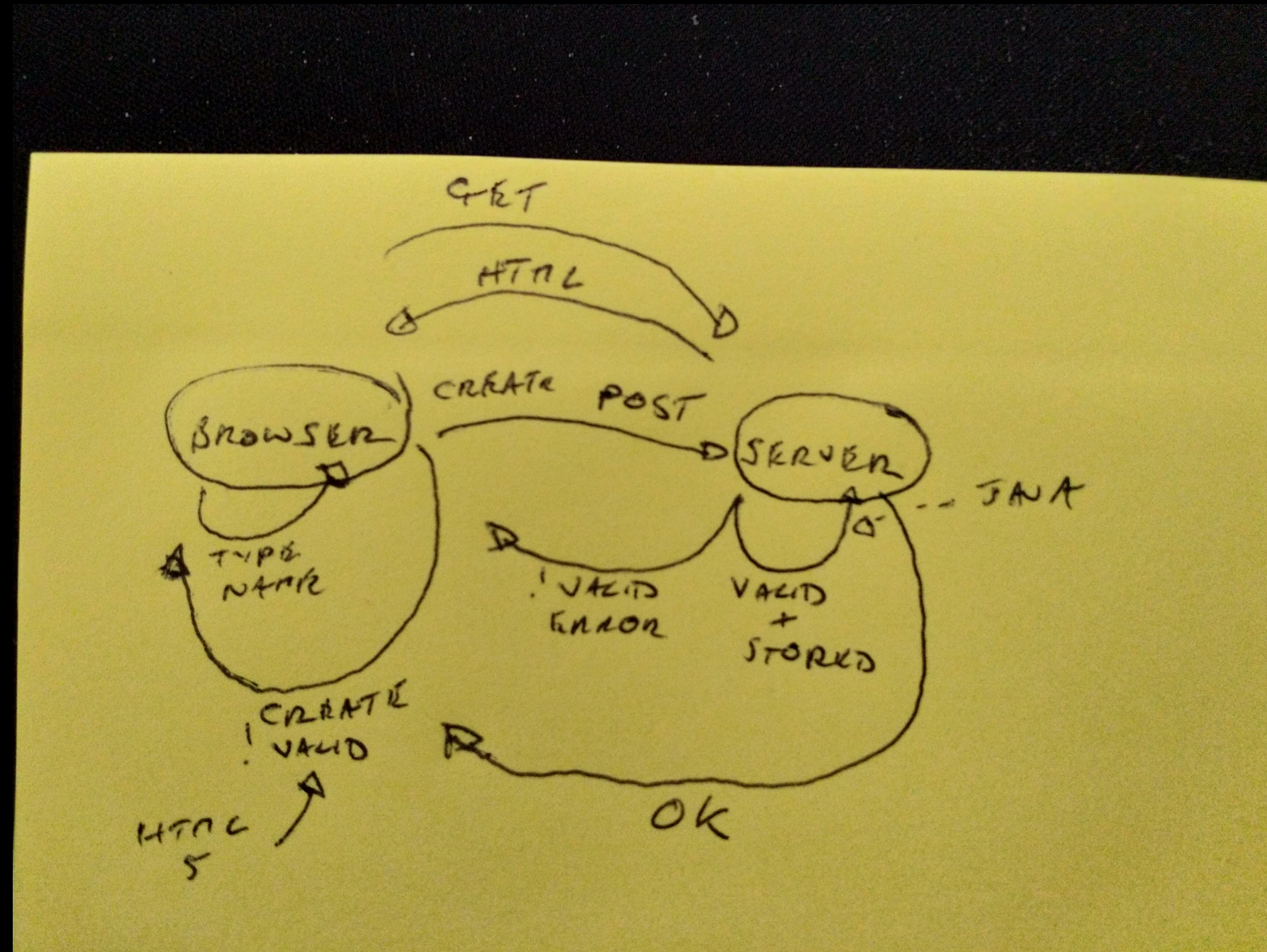
**Then someone
answers our
questions**

**And our world explodes with
detail**

Uhhh, OK

*"The validation rules are controlled by HTML using HTML5 **required** and **pattern** attributes. When the **[Create]** button is pressed an **HTTP POST** request is made to the server with the **form** details **x-www-form-urlencoded**. The author is stored in memory with an automatically generated id."*

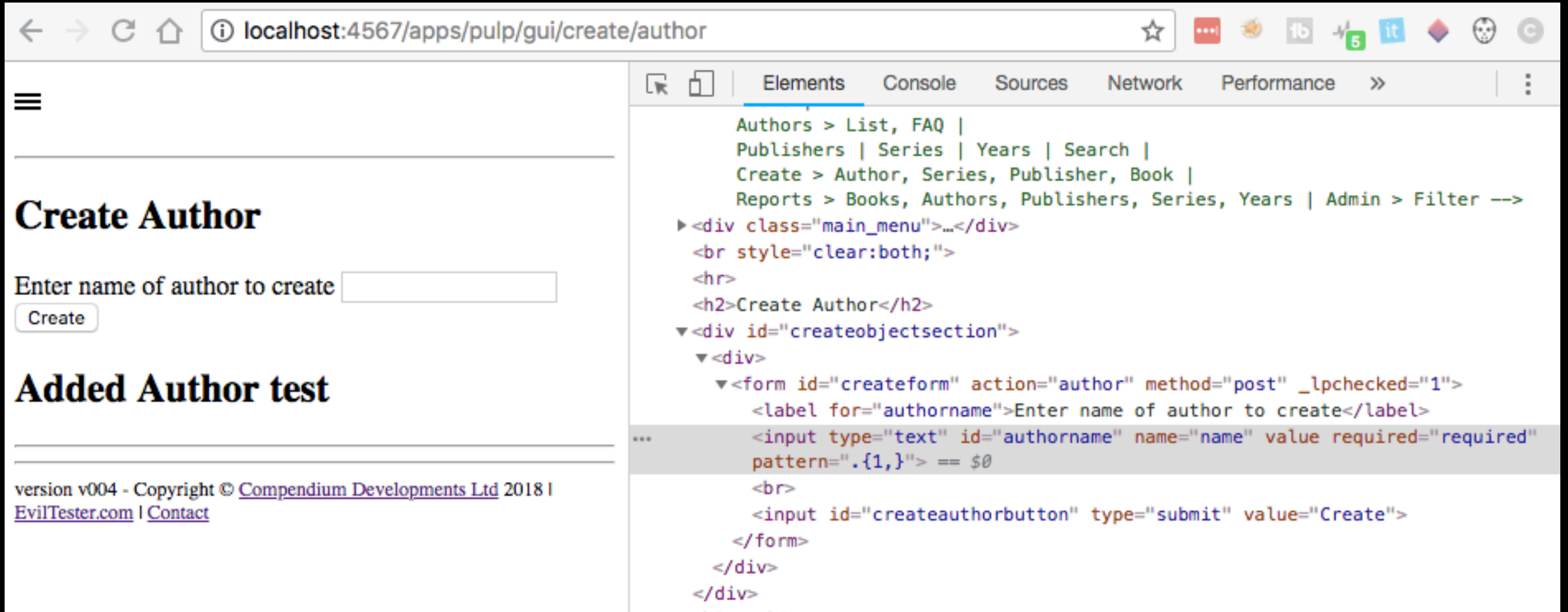
Maybe a diagram would help?



Now I have to learn about

- HTML, HTML5
- required
- pattern
- HTTP
- POST
- form, form encoding

And how do I 'see' the HTML?



The screenshot shows a web browser at the URL `localhost:4567/apps/pulp/gui/create/author`. The page displays a 'Create Author' form with a text input field and a 'Create' button. Below the form, it says 'Added Author test'. The browser's developer tools are open, showing the 'Elements' tab. The HTML structure is as follows:

```
Authors > List, FAQ |
Publishers | Series | Years | Search |
Create > Author, Series, Publisher, Book |
Reports > Books, Authors, Publishers, Series, Years | Admin > Filter -->
<div class="main_menu">...</div>
<br style="clear:both;">
<hr>
<h2>Create Author</h2>
<div id="createobjectsection">
  <div>
    <form id="createform" action="author" method="post" _lpchecked="1">
      <label for="authorname">Enter name of author to create</label>
      <input type="text" id="authorname" name="name" value required="required"
        pattern="{1,}"> == $0
      <br>
      <input id="createauthorbutton" type="submit" value="Create">
    </form>
  </div>
</div>
```

OMG I can change it too

```
▼<div>
  ▼<form id="createform" action="author" method="post">
    <label for="authorname">Enter name of author to create</label>
    <input type="text" id="authorname" name="name" value="" required="" pattern=
    {1,}"> == $0
    <br>
    <input id="createauthorbutton" type="submit" value="Create">
  </form>
</div>
```

What if I change the form so there is no validation?

Then the server side validation should catch that.



Create Author

Enter name of author to create

Create

Error: Could not find a name to add

So now I have to test the server side validation as well?

"That will take quite a lot of time amending the form each time."

"Test it at the HTTP level then"

"How?"

"Use an HTTP Debug Proxy."

"A Wha?"

An HTTP Debug Proxy

Id	Req. Timestamp	Method	URL	Code	Reason	RTT	Size Resp. Body	Highest Alert	Note	Tags
1	13/09/18 11:47:06	GET	http://localhost:4567/apps/pulp/gui/create/author	200	OK	11 ms	5,863 bytes	🟡 Medium		Form, Script, Comment
7	13/09/18 11:47:06	GET	http://localhost:4567/apps/pulp/v003/css/app.css	200	OK	3 ms	30 bytes	🟡 Low		
8	13/09/18 11:47:06	GET	http://localhost:4567/apps/pulp/v003/css/dropdo...	200	OK	3 ms	1,889 bytes	🟡 Low		Comment
10	13/09/18 11:47:06	GET	http://localhost:4567/apps/pulp/v003/js/confirmati...	200	OK	5 ms	1,365 bytes	🟡 Low		Comment
23	13/09/18 11:47:31	POST	http://localhost:4567/apps/pulp/gui/create/author	200	OK	6 ms	5,939 bytes	🟡 Medium		Form, Script, Comment
24	13/09/18 11:47:31	GET	http://localhost:4567/apps/pulp/v003/css/app.css	200	OK	3 ms	30 bytes	🟡 Low		
25	13/09/18 11:47:31	GET	http://localhost:4567/apps/pulp/v003/js/confirmati...	200	OK	3 ms	1,365 bytes	🟡 Low		Comment
26	13/09/18 11:47:31	GET	http://localhost:4567/apps/pulp/v003/css/dropdo...	200	OK	3 ms	1,889 bytes	🟡 Low		Comment

What Questions Did I Ask?

- How does this work?
- What does this do?
- How can I see that? How can I make that happen?
- What should it do?
- When should it do that?
- Where should I see that?

How?
Not Why?

What is Hard About This?

Why do people find this tough?

- don't like to be seen as 'not knowing' something
- don't like to say "I don't know"

Let's see, what could we be?

- Scientist
- Skeptic
- Believer
- Seeker



Ken Campbell

"Seeker!"

Robert Anton Wilson said...

"Belief is the death of intelligence. As soon as one believes a doctrine of any sort, or assumes certitude, one stops thinking about that aspect of existence."

Robert Anton Wilson, Cosmic Trigger

Top Three Trouble-free Tactics To Tough Talk

- Knowing nothing about something is good
- You don't know nothing about it, what you know about it is nothing
- Ask questions when you do know stuff

I know nothing about

- Kardashians
- Love Island
- Big Brother

That is good.

You don't know nothing about it

What you know about it, you describe as 'nothing'

- You ask questions to see if that description fits
- You try to expand your model
- You try to invalidate parts of your model

Same as normal

As a Manager

I ask questions when I do know stuff because:

- I know other people in the room don't know
- I know people have different understandings
- It gives the speaker a chance to show their knowledge

**Asking questions is
good**

The Tester!



Asking Questions is Good

End Credits

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BIO

Alan is a test consultant who enjoys testing at a technical level using techniques from psychotherapy and computer science. In his spare time Alan is currently programming a multi-user text adventure game and some buggy JavaScript games in the style of the Cascade Cassette 50. Alan is the author of the books "Dear Evil Tester", "Java For Testers" and "Automating and Testing a REST API".

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Alan's main website is compendiumdev.co.uk and he blogs at blog.eviltester.com